Materials Engineering, BSMSE

Location(s): West Lafayette

About the Program

The BS program in Materials Science and Engineering is accredited by the Engineering Accreditation Commission of ABET, under the commission's General Criteria and Program Criteria for Materials, Metallurgical, Ceramics and Similarly Named Engineering Programs.

Materials have enabled improvement in the products humans use since the beginning of recorded history. For example, without the development of high purity silicon most of today's electronic devices would not exist. New low density, high stiffness composite materials have replaced metals and wood in tennis racquets. The performance of these materials stems from their properties which depend on their microscopic structure, also known as microstructure. Microstructure in turn depends upon materials processing, the fabrication of materials into functional shapes. Materials Engineering is the study of the interrelationships between processing, structure, properties and performance of materials.

Materials engineers study the structure and composition of materials on scales ranging from the electronic and atomic through the microscopic to the macroscopic. They develop new materials, improve traditional materials and are key people in the manufacturing process to produce materials reliably and economically. They seek to understand phenomena and to measure materials properties of all kinds, and they predict and evaluate the performance of real materials as structural or functional elements in engineering systems. Employment opportunities span all types of industry, such as aerospace, automotive, chemical, electronic, energy and primary material-producing companies.

Materials Engineering

Materials Engineering Major Change (CODO) Requirements

Degree Requirements

125 Credits Required

Materials Engineering Major Requirements (63 credits)

Materials Engineering Major Courses (43 credits)

A GPA of 2.0 is required in MSE 20000 and 30000 level courses and MSE 42000.

- MSE 23000 - Structure And Properties Of Materials **Credit Hours: 3.00**
- MSE 23500 - Second Year Materials Engineering Laboratory **Credit Hours: 3.00**
- MSE 25000 - Physical Properties In Engineering Systems **Credit Hours: 3.00**
- MSE 26000 - Thermodynamics Of Materials **Credit Hours: 3.00**

- MSE 27000 - Atomistic Materials Science **Credit Hours: 3.00**
- MSE 33000 - Processing And Properties Of Materials **Credit Hours: 3.00**
- MSE 33500 - Third Year Materials Engineering Laboratory **Credit Hours: 3.00**
- MSE 34000 - Transport Phenomena **Credit Hours: 3.00**
- MSE 37000 - Electrical, Optical, And Magnetic Properties Of Materials **Credit Hours: 3.00**
- MSE 38200 - Mechanical Response Of Materials **Credit Hours: 3.00**
- MSE 39000 - Materials Engineering Seminar **Credit Hours: 0.00** - must be passed with a grade of S (Satisfactory)
- MSE 39100 - Materials Engineering Seminar II **Credit Hours: 1.00**
- MSE 42000 - Structure And Properties Of Organic Materials **Credit Hours: 3.00**
- MSE 43000 - Materials Processing And Design I **Credit Hours: 3.00**
- MSE 44000 - Materials Processing And Design II **Credit Hours: 3.00**
- MSE 44500 - Materials Engineering Systems Analysis And Design **Credit Hours: 3.00**

MSE Technical and Support Area Selectives (20 credits)

- Technical Selectives and Support Area Selectives - Credit Hours: 20.00 (see Materials Engineering Supplemental Information for details on meeting this requirement)

Other Departmental/Program Course Requirements (62-70 credits)

Other Course Requirements (38-45 credits)

Intro to Engineering I & II - Credit Hours: 4.00-8.00

- ENGR 13100 - Transforming Ideas To Innovation I (UCC: AI/IL) **Credit Hours: 2.00**

  and
- ENGR 13200 - Transforming Ideas To Innovation II (UCC: AI) **Credit Hours: 2.00**

  OR
- EPCS 11100 - First Year Participation In EPICS I (UCC: STS) **Credit Hours: 1.00** and
- EPCS 12100 - First Year Participation In EPICS II (UCC: STS) **Credit Hours: 1.00** and
- ENGR 13300 - Transforming Ideas To Innovation, EPICS/VIP (UCC: AI/IL) **Credit Hours: 2.00**

  OR
- VIP 17911 - First Year Participation In Vertically Integrated Projects (VIP) I (UCC: STS) **Credit Hours: 1.00** and
- VIP 17912 - First Year Participation In Vertically Integrated Projects (VIP) II **Credit Hours: 1.00** and
- ENGR 13300 - Transforming Ideas To Innovation, EPICS/VIP (UCC: AI/IL) - **2.00 Credit Hours**

  OR
- ENGR 16100 - Honors Introduction To Innovation And The Physical Science Of Engineering Design I (UCC: AI/IL) **Credit Hours: 4.00** and
- ENGR 16200 - Honors Introduction To Innovation And The Physical Science Of Engineering Design II (UCC: SCI) **Credit Hours: 4.00**

  OR
- ENGR 13000 - Transforming Ideas Into Innovations (UCC: AI/IL) **Credit Hours: 4.00**

- CHM 11510 - General Chemistry I **Credit Hours: 3.00**

  AND
- CHM 11520 - General Chemistry I - Laboratory (UCC: SCI) **Credit Hours: 1.00** or
- CHM 11530 - General Chemistry I - Virtual Laboratory (UCC: SCI) **Credit Hours: 1.00**

- CHM 11610 - General Chemistry II **Credit Hours: 3.00** (C- or better)
- CHM 11620 - General Chemistry II - Laboratory (UCC: SCI) **Credit Hours: 1.00** (C- or better)
- MA 26100 - Multivariate Calculus (UCC: QR) **Credit Hours: 4.00**

- MA 16100 - Plane Analytic Geometry And Calculus I (UCC: QR) **Credit Hours: 5.00** or
- MA 16500 - Analytic Geometry And Calculus I (UCC: QR) **Credit Hours: 4.00**

- MA 16200 - Plane Analytic Geometry And Calculus II (UCC: QR) **Credit Hours: 5.00** or
- MA 16600 - Analytic Geometry And Calculus II (UCC: QR) **Credit Hours: 4.00**

Mathematics Linear & Differential Equations Requirement - Credit Hours: 6.00-7.00

- MA 26500 - Linear Algebra (UCC: QR) **Credit Hours: 3.00** and
- MA 26600 - Ordinary Differential Equations (UCC: QR) **Credit Hours: 3.00**
  OR
- MA 26200 - Linear Algebra And Differential Equations (UCC: QR) **Credit Hours: 4.00**
  AND
- MA 30300 - Differential Equations And Partial Differential Equations For Engineering And The Sciences **Credit Hours: 3.00** or
- MA 35100 - Elementary Linear Algebra (UCC: QR) **Credit Hours: 3.00**

Physics I - Credit Hours: 4.00

- PHYS 17200 - Modern Mechanics (UCC: SCI) **Credit Hours: 4.00**
  OR
- ENGR 16100 - Honors Introduction To Innovation And The Physical Science Of Engineering Design I (UCC: AI/IL) - **4.00 Credit Hours** and
- ENGR 16200 - Honors Introduction To Innovation And The Physical Science Of Engineering Design II (UCC: SCI) - **4.00 Credit Hours**

Physics II - Credit Hours: 0.00-4.00

- PHYS 24100 - Electricity And Optics (UCC: SCI) **Credit Hours: 3.00** ♦ and
- PHYS 25200 - Electricity And Optics Laboratory **Credit Hours: 1.00** ♦
  OR
- PHYS 27200 - Electric And Magnetic Interactions (UCC: SCI) **Credit Hours: 4.00** ♦

Engineering General Education Requirements (24-25 credits)

For a list of courses that satisfy this requirement, see Supplemental Information

- General Education I (UCC: HUM) - Credit Hours: 3.00-4.00
- General Education II (UCC: BSS) - Credit Hours: 3.00
- General Education III - Credit Hours: 3.00
- General Education IV - Credit Hours: 3.00
- General Education V - Credit Hours: 3.00
- General Education VI - Credit Hours: 3.00
- Written Communication (UCC: WC) - Credit Hours: 3.00-4.00
- Oral Communication (UCC: OC) - Credit Hours: 3.00

Additional General Education Requirements

- At least 6 credits must be in non-introductory courses (30000-level or above OR from courses with a required pre-requisite in the same department)
- At least 12 credits must be taken from the College of Liberal Arts, the Daniels School of Business, and/or the Honors College provided such courses are not focused primarily on engineering, technology, the natural sciences, or mathematics.
- A course is not eligible for use towards the 24 credits of Engineering General Education requirement if that course is used to satisfy a different engineering degree requirement.

Supplemental Information

Click here for Materials Engineering Supplemental Information

Optional Concentration

- Semiconductors and Microelectronics Concentration for Materials Science Engineering

GPA Requirements

- A GPA of 2.0 is required for all Baccalaureate degrees.
- A GPA of 2.0 is required in MSE 20000 and 30000 level courses and MSE 42000.

Pass/No Pass Policy

- Courses with a Pass/No Pass grade (P/NP) may be used for General Education courses only. Courses taken to satisfy UCC requirements must still be taken for a grade, even if also used as a General Education course.

Transfer Credit Policy

If you are interested in registering for a course offered by a different institution, please review the information on the Purdue Office of the Registrar Transfer Credit website.
* If you are an admitted transfer student, please review the above Transfer Credit website and then work with your academic advisor to determine exactly how your previous courses might apply.

Course Requirements and Notes

- MSE 39000 must be taken once and successful passed with a grade of S(Satisfactory) before taking MSE 39100.

University Requirements

University Core Requirements

FOR A COMPLETE LISTING OF UNIVERSITY CORE COURSE SELECTIVES, VISIT THE UNIVERSITY SENATE WEBSITE.

- AI Working Competency (UCC: AI)
- Human Cultures: Behavioral/Social Science (UCC: BSS)
- Human Cultures: Humanities (UCC: HUM)
- Information Literacy (UCC: IL)
- Oral Communication (UCC: OC)

- Quantitative Reasoning (UCC: QR)
- Science #1 (UCC: SCI)
- Science #2 (UCC: SCI)
- Science, Technology, and Society (UCC: STS)
- Written Communication (UCC: WC)

Civics Literacy Proficiency Requirement

The Civics Literacy initiative is designed to develop civic knowledge of Purdue students to graduate a more informed citizenry. For more information visit the Civics Literacy - Office of the Vice Provost for Teaching and Learning - Purdue University.

To obtain the Civics Literacy credential, all Baccalaureate degree seeking students will complete the Civics Literacy Podcast pathway - Complete 12 podcasts created by the Purdue Center for C-SPAN Scholarship & Engagement that use C-SPAN material and pass the required quizzes per episode. After the pathway has been successfully completed, students will take the Civics Literacy knowledge test. Civics Literacy does not impact a student's GPA or bear any financial costs.

More details about each pathway and how to complete the requirement can be found on the Civics Literacy Student Dashboard in myPurdue.

Upper Level Requirement

- Resident study at Purdue University for at least two semesters and the enrollment in and completion of at least 32 semester hours of coursework required and approved for the completion of the degree. These courses are expected to be at least junior-level (30000+) courses.
- Students should be able to fulfill *most*, *if not all*, of these credits within their major requirements; there should be a clear pathway for students to complete any credits not completed within their major.

Sample Plan of Study

The following sample plan of study is a suggested arrangement of courses. Consultation with an advisor may result in an altered plan customized for an individual student.

Fall 1st Year

- Intro to Engineering I – Credit Hours: 2.00-4.00
- Written Communication (UCC: WC) or Oral Communication (UCC:OC) - Credit Hours: 3.00-4.00
- CHM 11510 - General Chemistry I **Credit Hours: 3.00**

- CHM 11520 - General Chemistry I - Laboratory (UCC: SCI) **Credit Hours: 1.00** or
- CHM 11530 - General Chemistry I - Virtual Laboratory (UCC: SCI) **Credit Hours: 1.00**

- MA 16100 - Plane Analytic Geometry And Calculus I (UCC: QR) **Credit Hours: 5.00** or
- MA 16500 - Analytic Geometry And Calculus I (UCC: QR) **Credit Hours: 4.00**

13-17 Credits

Spring 1st Year

- Intro to Engineering II - Credit Hours: 2.00-4.00

- Written Communication (UCC: WC) or Oral Communication (UCC:OC) - Credit Hours: 3.00-4.00
- Physics I - Credit Hours: 4.00
- CHM 11610 - General Chemistry II **Credit Hours: 3.00**
- CHM 11620 - General Chemistry II - Laboratory (UCC: SCI) **Credit Hours: 1.00**

- MA 16200 - Plane Analytic Geometry And Calculus II (UCC: QR) **Credit Hours: 5.00** or
- MA 16600 - Analytic Geometry And Calculus II (UCC: QR) **Credit Hours: 4.00**

17-21 Credits

Fall 2nd Year

- MA 26100 - Multivariate Calculus (UCC: QR) **Credit Hours: 4.00**
- MSE 23000 - Structure And Properties Of Materials **Credit Hours: 3.00**
- MSE 27000 - Atomistic Materials Science **Credit Hours: 3.00**
- MSE 25000 - Physical Properties In Engineering Systems **Credit Hours: 3.00**
- MSE 39000 - Materials Engineering Seminar **Credit Hours: 0.00**
- Mathematics Linear & Differential Equations Requirement - Credit Hours: 3.00

16 Credits

Spring 2nd Year

- MSE 23500 - Second Year Materials Engineering Laboratory **Credit Hours: 3.00**
- MSE 26000 - Thermodynamics Of Materials **Credit Hours: 3.00**
- Mathematics Linear & Differential Equations Requirement - Credit Hours: 3.00-4.00
- Physics II - Credit Hours: 4.00
- General Education I (UCC: HUM) - Credit Hours: 3.00-4.00

16-17 Credits

Fall 3rd Year

- MSE 33500 - Third Year Materials Engineering Laboratory **Credit Hours: 3.00**
- MSE 34000 - Transport Phenomena **Credit Hours: 3.00**
- MSE 37000 - Electrical, Optical, And Magnetic Properties Of Materials **Credit Hours: 3.00**
- MSE 39100 - Materials Engineering Seminar II **Credit Hours: 1.00**
- MSE 42000 - Structure And Properties Of Organic Materials **Credit Hours: 3.00**
- General Education II (UCC: BSS) - Credit Hours: 3.00

16 Credits

Spring 3rd Year

- MSE 33000 - Processing And Properties Of Materials **Credit Hours: 3.00**
- MSE 38200 - Mechanical Response Of Materials **Credit Hours: 3.00**
- General Education III - Credit Hours: 3.00

- Technical Selective I - Credit Hours: 3.00
- Technical Selective II - Credit Hours: 3.00

15 Credits

Fall 4th Year

- MSE 43000 - Materials Processing And Design I **Credit Hours: 3.00**
- MSE 44500 - Materials Engineering Systems Analysis And Design **Credit Hours: 3.00**
- General Elective IV - Credit Hours: 3.00
- Technical Selective III - Credit Hours: 3.00
- Technical Selective IV - Credit Hours: 3.00

15 Credits

Spring 4th Year

- MSE 44000 - Materials Processing And Design II **Credit Hours: 3.00**
- General Education V - Credit Hours: 3.00
- General Education VI - Credit Hours: 3.00
- Technical Selective V or Support Area Selective I - Credit Hours: 3.00
- Technical Selective VI or Support Area Selective II - Credit Hours: 3.00
- Technical Selective VII or Support Area Selective III - Credit Hours: 2.00

17 Credits

Pre-Requisite Information

For pre-requisite information, log in to mypurdue.purdue.edu and click here.

World Language Courses

World Language proficiency requirements vary by program. The following list is inclusive of all world languages PWL offers for credit; for acceptable languages and proficiency levels, see your advisor. (ASL-American Sign Language; ARAB-Arabic; CHNS-Chinese; FR-French; GER-German; GREK-Greek(Ancient); HEBR-Hebrew(Biblical); HEBR-Hebrew(Modern); ITAL-Italian; JPNS-Japanese; KOR-Korean; LATN-Latin; PTGS-Portuguese; RUSS-Russian; SPAN-Spanish)

Critical Course

The ♦ course is considered critical.

In alignment with the Degree Map Guidance for Indiana's Public Colleges and Universities, published by the Commission for Higher Education (pursuant to HEA 1348-2013), a Critical Course is identified as "one that a student must be able to pass to persist and succeed in a particular major. Students who want to be nurses, for example, should know that they are expected to be proficient in courses like biology in order to be successful. These would be identified by the institutions for each degree program."

Disclaimer

The student is ultimately responsible for knowing and completing all degree requirements. Consultation with an advisor may result in an altered plan customized for an individual student. The myPurduePlan powered by DegreeWorks is the knowledge source for specific requirements and completion.

Comparative information about Purdue University and other U.S. educational institutions is also available through the College Navigator tool, provided by the National Center for Education Statistics, and through the U.S. Department of Education College Scorecard.